MEDIUM

ADVANCED MATH

Nonlinear functions

07

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

The function f is defined by $f x = 4x^{-1}$. What is the value of $f 21$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$

Q2 ADVANCED MATH

In the xy-plane, the y-coordinate of the y-intercept of the graph of the function f is c. Which of the following must be equal to c ?

A. $f(0)$

B. $f(1)$

C. $f(2)$

D. $f(3)$

Q3

ADVANCED MATH

$$fx = x^2 - 18x - 360$$

If the given function f is graphed in the xy-plane, where $y = fx$, what is an x-intercept of the graph?

A. $-12, 0$

B. $-30, 0$

C. $-360, 0$

D. $12, 0$

Q4 ADVANCED MATH

A physics class is planning an experiment about a toy rocket. The equation $y = -16x - 5.6^2 + 502$ gives the estimated height y, in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy-plane?

A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.

B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.

D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

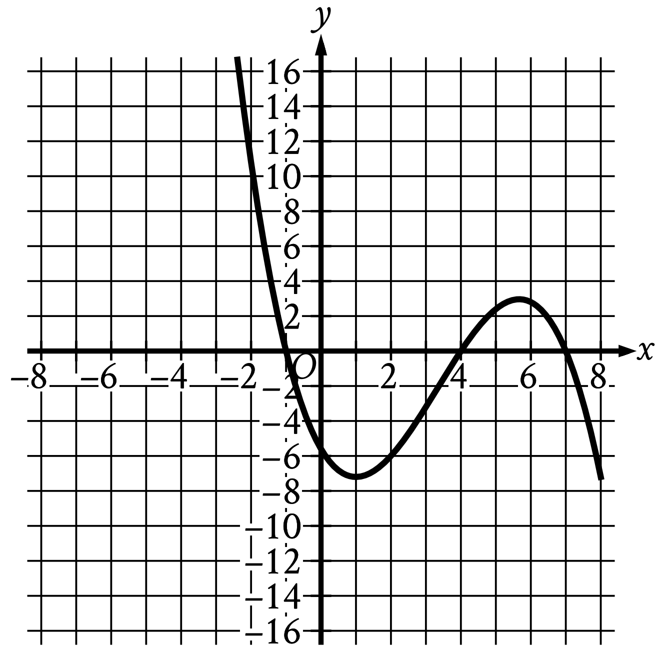
Q5 GRID-IN ADVANCED MATH

The function f is defined by $f x = 5\frac{1}{4} - x^{2} + \frac{11}{4}$. What is the value of $f\frac{1}{4}$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- In quadrant 2 the curve falls sharply to cross the x axis at point (negative 1 comma 0).
- In quadrant 3 the curve falls sharply to cross the y axis at point (0 comma negative 5.5).
- In quadrant 4:
  - The curve falls gradually to a relative minimum at point (1 comma negative 7).
  - The curve rises sharply to cross the x axis at point (4 comma 0).
- In quadrant 1:
  - The curve rises sharply to a relative maximum at point (5.5 comma 3).
  - The curve falls sharply to cross the x axis at point (7 comma 0).
- In quadrant 4 the curve falls sharply.

The graph of $y = fx$ is shown, where the function f is defined by $fx = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x

does $fx = 0$?

A. One

B. Two

C. Three

D. Four

Q7 ADVANCED MATH

A rectangle has a length of x units and a width of x - 15 units. If the rectangle has an area of 76 square units, what is the value of x?

A. 4

B. 19

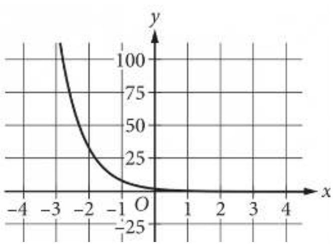
C. 23

D. 76

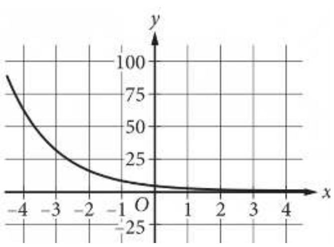
$$y = 4(2^x)$$

Which of the following is the graph in the xy-plane of the given equation?

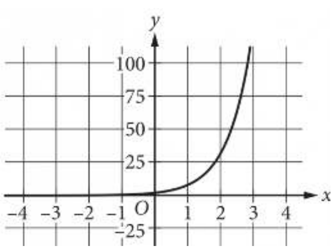
A.



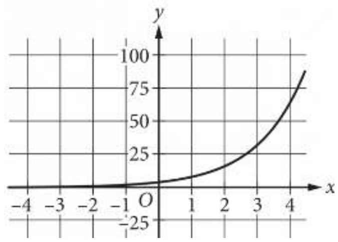
B.



C.



D.


y
100
75
50
25
−25
x
−4
−3
−2
−1
O
1
2
3
4

Q9

ADVANCED MATH

The height, in feet, of an object x seconds after it is thrown straight up in the air can be modeled by the function $h(x) = -16x^2 + 20x + 5$. Based on the model, which of the following statements best interprets the equation $h(1.4) = 1.64$?

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

B. The height of the object 1.64 seconds after being thrown straight up in the air is 1.4 feet.

C. The height of the object 1.64 seconds after being thrown straight up in the air is approximately 1.4 times as great as its initial height.

D. The speed of the object 1.4 seconds after being thrown straight up in the air is approximately 1.64 feet per second.

Q10 ADVANCED MATH

A company opens an account with an initial balance of $ 36,100.00. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A, where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is $ 68,071.93. Which equation could define A?

A. $At = 36{,}100.001.05^t$

B. $At = 31{,}971.931.05^t$

C. $At = 31{,}971.930.05^t$

D. $At = 36{,}100.000.05^t$